

SignalScope Integration Report

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Scope

This report summarizes the current end-to-end integration between the Django backend, the PyTorch/ResNet18 inference path, and the Next.js frontend.

Connected Paths

- Frontend calls **GET /api/ping/** to verify backend availability on load.
- Frontend submits multipart image uploads to **POST /api/predict/** and renders the returned verdict and probabilities.
- Django view code forwards image bytes into **detector.services.classify_image**, which loads a checkpointed PyTorch ResNet18 model when available.
- The explain endpoint reuses the same inference path and appends safe limitation text for development use.

Validation Results

Check	Result
pytest backend/detector/tests/test_api.py tests/test_ml_pipeline.py	6 passed
cd frontend && npm run build	Succeeded

Current Limitation

The repository is wired for production-style checkpoint loading, but a trained checkpoint is still required for meaningful detection quality. Without a trained checkpoint, the integration remains a development baseline.

Primary Files

- backend/detector/services.py
- backend/detector/views.py
- frontend/src/app/page.tsx
- tests/test_ml_pipeline.py